EXALT Technologies

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Smart Multi-Organization Appointment Booking & Management Platform

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Table Of Contents

1. INTRODUCTION 1

2. PRODUCT VISION 2

2.1 Customer Problems 2

2.2 Organization Problems 3

2.3 Platform Problems 3

3. PROPOSED SOLUTION 3

3.1 SYSTEM USERS 4

3.1.1 Super Admin 4

3.1.2 Organization 5

3.1.3 Organization Admin 5

3.1.4 CRM 5

3.1.5 Worker 6

3.1.6 Customer 6

3.2 CORE PRODUCT FEATURES 7

3.2.1 Location-Based Organization Discovery 7

3.2.2 Organization Profile and Location Management 7

3.2.3 Service Catalog Management 8

3.2.4 Room and Resource Management 8

3.2.5 Company Working Hours, Special Days, Worker Schedules, and Real-Time Worker Status 8

3.2.6 Dynamic Availability Calculation and Assignment 10

3.2.7 Appointment Booking Flow 10

3.2.8 Appointment Lifecycle Management 11

3.2.9 Appointment Calendar Management 12

3.2.10 Appointment History and Audit Trail 12

3.2.11 Worker Block Requests 12

3.2.12 Payment Management 13

3.2.13 Reviews and Ratings 13

3.2.14 Favorites 13
3.2.15 Notification System 13
3.2.16 System Messages and Announcements 14
3.2.17 Role-Based Access Control (RBAC) 14
3.3 NON-FUNCTIONAL REQUIREMENTS 14
3.3.1 Security 15
3.3.2 Performance 15
3.3.3 Reliability 15
3.3.4 Scalability 15

1. INTRODUCTION

The Smart Multi-Organization Appointment Booking & Management Platform is a multi-organization appointment marketplace that connects customers with service providers such as clinics, salons, consulting offices, repair centers, educational providers, and other appointment-based businesses.

The platform enables organizations to create professional profiles, publish their services, define their availability, manage their internal staff and physical resources, and handle customer appointments. Customers can discover organizations based on services and geographical location, view available appointment times, reserve appointments, complete payments, and provide feedback through ratings and reviews.

It also includes a Super Admin management layer responsible for governing the marketplace, monitoring organizations and users, managing platform permissions, tracking revenue, and ensuring operational reliability.

Each side of the marketplace has different objectives. Customers need a simple and trustworthy way to discover suitable organizations, compare options, and reserve appointments. Organizations need tools to manage schedules, staff, services, physical resources, customers, and revenue while reducing manual workload. Platform administration needs complete visibility and control over marketplace activity, security, quality, and growth.

2. PRODUCT VISION

The goal of the platform is to become a centralized digital marketplace where customers can easily discover and book services while organizations can efficiently manage their appointment operations.

The platform transforms traditional appointment processes based on phone calls, manual calendars, spreadsheets, and separate payment systems into an automated ecosystem providing real-time availability, location-based discovery, digital appointment management, integrated payment tracking, customer engagement, and marketplace analytics.

2.1 Customer Problems

Customers currently face several difficulties when booking services. Finding suitable service providers requires manual searching, and it is difficult to compare organizations based on location, price, availability, and reputation all at once. Customers cannot always see available appointment times before contacting the organization directly, which means booking often requires unnecessary back-and-forth communication through phone calls. On top of this, customers have limited visibility into their own appointment history and payment records, and there is no unified place where they can save preferred organizations and keep track of upcoming appointments.

2.2 Organization Problems

Organizations experience their own set of operational challenges. Appointment management is often manual and error-prone, made more complicated by the fact that staff members may have different availability schedules from one another and from the organization as a whole, and that not every worker can perform every service or use every physical resource. This combination can lead to double-booking and scheduling conflicts, while customer information ends up scattered across different systems rather than living in one place. Managing holidays, special working days, and blocked periods adds further difficulty, and payment tracking remains disconnected from appointment records. On top of all this, organizations lack the analytics they would need to understand their customers, appointments, and overall performance.

2.3 Platform Problems

Without centralized management, platform operators are left without the tools they need to run a healthy marketplace. They cannot monitor organization activity, control marketplace quality, or manage permissions effectively. They also lack the ability to track revenue, detect suspicious activity, or analyze marketplace growth, leaving them with little real insight into how the platform is performing.

3. PROPOSED SOLUTION

The platform provides a complete appointment ecosystem serving three groups of users. Organizations can register and create a business profile, publish the services they offer along with prices and standard durations, and manage their internal staff and physical resources so that available appointment times are calculated automatically. From there, organizations handle customer appointments as they come in, track payments against those appointments, and receive customer reviews once services are delivered. Customers, for their part, can create personal accounts and use them to discover organizations nearby, searching for services by category or location. They can view organization profiles, compare ratings and availability side by side, and reserve appointments that fit their schedule. Once booked, customers can manage their upcoming appointments, track payment information, add organizations they like to their favorites, and submit reviews after their visit. Super Admins sit above both groups, managing organizations by approving or suspending businesses as needed, monitoring all appointments across the platform, reviewing system activity, and analyzing overall marketplace performance.

Underlying all of this, every authenticated person on the platform is stored as a single Person entity, which acts as the shared identity layer for everyone who uses the system. Depending on their responsibilities, a person may also carry a customer profile or a staff profile tied to an organization. Separately from these profile extensions, the platform recognizes a fixed set of predefined roles that are assigned to people and enforced entirely by the application's internal logic rather than by database-level permission tables: Super Admin, Organization Admin, CRM (also referred to as Manager), Worker, and Customer. These roles are defined by the developer and cannot be created, modified, or deleted by organizations, and each carries a fixed set of permissions enforced consistently across the whole platform.

3.1 SYSTEM USERS

3.1.1 Super Admin

The Super Admin represents the platform operator responsible for controlling and monitoring the entire marketplace, and is not modeled as a separate entity in the data model; rather, the Super Admin is simply a person who has been assigned the Super Admin role and receives their capabilities entirely through that role's permissions. In this role, the Super Admin approves new organizations before they go live and suspends any organization found to be violating platform policies. They manage users and organizations across the system, monitor appointments as they happen, analyze revenue, and track overall platform activity. To support this, the Super Admin dashboard surfaces key business metrics: the number of organizations on the platform, the number of active customers, appointment volume, revenue generated, organization performance, and general user activity, giving the Super Admin a full picture of marketplace health at a glance.

3.1.2 Organization

An organization is a registered business that provides services through the platform, such as a medical clinic, a beauty salon, a consulting office, or a repair center. Each organization's profile carries a name, an email address, a phone number, a description, a profile picture, a single physical location, a status, and a creation date. In the current version of the platform, an organization has exactly one location; branches and multiple locations are intentionally out of scope, though the underlying design remains extensible enough to support that in a future version without a fundamental rework, for now, if an organization need to have different branches it must have an independent account for each. Organizations move through a controlled lifecycle as they join the platform: they begin in a Pending Approval state, move to Active once approved, and can later be Suspended or Removed if necessary, with the Super Admin controlling this approval and status at every stage. Operationally, the organization is not managed by a single undifferentiated group of staff; instead, its internal work is divided across three distinct roles, described below, each with a clearly scoped set of responsibilities.

3.1.3 Organization Admin

The Organization Admin is the highest-level user within a given organization and is responsible for the business as a whole from an administrative standpoint. This role manages the organization's profile, service catalog, physical resources, working schedules, and special days, and is responsible for adding and managing the organization's staff by assigning each person the appropriate internal role, whether CRM or Worker. The Organization Admin also has visibility into the organization's overall performance, including appointment volume, revenue, and reviews, and serves as the primary point of accountability that the Super Admin interacts with when approving, suspending, or otherwise governing the organization. In practice, the Organization Admin sets up and maintains the structure that the CRM and Worker roles then operate within on a day-to-day basis.

3.1.4 CRM

The CRM role, also referred to as Manager, is responsible for the day-to-day coordination of appointments within an organization. Rather than performing the services themselves, CRM users act as dispatchers who keep the organization's schedule running smoothly. They review incoming appointment requests and approve them before they are confirmed, and where a customer has not chosen a specific Worker, the CRM confirms the Worker the system has assigned, adjusting the assignment manually if needed. CRM users are also responsible for reviewing and approving block requests submitted by Workers, meaning that when a Worker asks to be marked unavailable for vacation, a personal matter, or any other reason, it is the CRM who evaluates and approves that request before it takes effect. Through these responsibilities, the CRM role effectively controls the flow of appointments from initial request through to confirmation, ensuring that no appointment goes unassigned and that no availability change happens without oversight.

3.1.5 Worker

Workers are individuals who actually perform appointment-based services on behalf of the organization, such as a doctor, a technician, a consultant, or a hair stylist. Each Worker profile includes a job title, a hiring date, a salary, a personal working schedule, and a real-time attendance status. Workers operate out of the organization's single business location rather than maintaining a location of their own, and they do not own any specific physical resource; instead, they move between whichever rooms or resources are appropriate and available for the service being performed. A Worker is not limited to a single service, either: each Worker is connected to the specific services they are qualified to perform through a many-to-many relationship, so that, for example, one Worker might be qualified for both haircuts and coloring while another is qualified only for massage. Once an appointment has been approved and assigned to them by the CRM, a Worker can view it on their personal schedule and carry out the service. At the start and end of each working day, a Worker also controls a simple real-time status toggle indicating whether they are currently checked in and available to be assigned new appointments, described further in Section 3.2.7. Workers can additionally submit requests to block off future time when they know in advance they will be unavailable, though these requests must be approved by the CRM before they affect the organization's published availability. After serving a customer, the Worker marks the appointment as Completed, which records both when the service actually finished and whether it finished earlier or later than scheduled.

3.1.6 Customer

Customers use the platform to discover organizations and reserve services. Their profile holds personal information such as birth date, alongside their appointments and their histories, their list of favorite organizations, and any reviews they have written. With this profile in place, customers can search for organizations, view providers near them, and browse the services those providers offer. They can reserve appointments and cancel them when plans change, and they can look back through their appointment history at any time. Customers can also track their payments, receive notifications about upcoming or changed appointments, and rate organizations appointments which reflects to its services and workers after their experience.

3.2 CORE PRODUCT FEATURES

3.2.1 Location-Based Organization Discovery

The platform enables customers to discover organizations based on geographical location. Customers can rely on their current location or search a different one entirely, and they can filter organizations by distance while viewing organization addresses directly in the results. Each search result surfaces the organization's name, its distance from the customer, its rating, its available appointment times, and the services it offers. For example, a customer searching for "Dental clinic near me" might see a result for ABC Dental Clinic showing a distance of 2 km, a rating of 4.8, and a next available slot tomorrow at 10:00.

3.2.2 Organization Profile and Location Management

Organizations must be able to create and maintain accurate business profiles so that customers understand their services, availability, and physical presence. Through their profile, organizations manage their business name, description, contact information, profile image, business status, and their single business location, which is what customers use when discovering the organization and where all of the organization's Workers are understood to operate from. The platform intentionally does not support multiple branches under one organization in its current version, nor does it track a separate physical location for individual Workers; every Worker belonging to an organization is assumed to work from that organization's one registered location. This keeps the location model simple while remaining structured in a way that could later support multiple locations without requiring a fundamental redesign.

3.2.3 Service Catalog Management

Organizations publish the services they provide through the platform, and each service carries a name, a description, a price, and a standard duration. Every service is connected to the specific Workers qualified to perform it and to the specific rooms or resources where it can take place, both through many-to-many relationships; a haircut, for example, might be performable by two different Workers and available in either of two different chairs. Customers can browse the organization's full service catalog, including which Workers offer a given service, before deciding on an appointment. A few business rules govern this catalog: inactive services cannot receive new appointments, the standard duration of a service determines how long its appointment reserves by default, prices are always displayed before a booking is confirmed, and organizations can update their services at any time without affecting appointments that have already been completed.

3.2.4 Room and Resource Management

Rather than tracking appointment availability as a set of pre-defined time slots, the platform models the physical spaces or equipment a service requires as rooms or resources, such as a hair chair, a doctor's examination room, or a consultation room. Each room belongs to a single organization and can support one or more services, and a given service may in turn be available in more than one room. Rooms are not tied to a specific Worker; different Workers may use the same room at different times, and a single Worker may use different rooms across the day depending on which one is free and appropriate for the service being performed. When an appointment is booked, the system assigns it to a specific room in addition to a specific Worker, and that assignment is stored as part of the final appointment record.

3.2.5 Company Working Hours, Special Days, Worker Schedules, and Real-Time Worker Status

Scheduling on the platform is built from several distinct layers that are deliberately kept separate rather than merged into one vague "availability" concept.

The foundation is the organization's working hours, which define the outer bounds of when the business operates at all-for example, Monday through Friday from 09:00 to 17:00. No appointment may ever be booked outside these hours, regardless of any individual Worker's own schedule. Organizations can also define special days that override these normal working hours for a specific date, covering situations such as holidays, temporary closures, or a day with different opening hours; a special day always takes priority over the organization's normal schedule for that date, and special days represent exceptions to the business's operating hours rather than an individual Worker's vacation or personal time off.

Sitting inside the organization's working hours is each Worker's own schedule, which defines when that particular Worker is normally expected to be working. If the organization is open from 09:00 to 17:00 but a given Worker's schedule only runs from 10:00 to 15:00, then bookable time for that Worker is limited to the overlap of the two, which in this example is 10:00 to 15:00. Booking is only ever permitted where the organization's hours and the Worker's own schedule both agree that time is open.

Separate again from either of these schedule layers is a real-time status that each Worker controls personally. This status simply answers the question of whether the Worker is currently checked in and present, similar to an attendance system rather than a scheduling one: the Worker switches it on when they arrive and off when they leave. To prevent a Worker from forgetting to switch it off, the system automatically forces the status to off the moment the organization's working hours end for the day, with no action required from the Worker. This real-time status affects same-day booking decisions - for instance, whether a Worker can be assigned an appointment starting in the next few minutes - but it is not a scheduling mechanism on its own and has no bearing on future days.

Because this real-time toggle only reflects the present moment, it does not replace the need for future block requests, described in Section 3.2.12. A Worker who knows in advance that they will be unavailable for a vacation, a meeting, a personal matter, or any other planned reason still submits a request specifying the future date and time range involved, and this request must be approved by the CRM before it affects the organization's published availability. In short, the real-time toggle answers "is this Worker here right now," while a block request answers "is this Worker planning to be unavailable later," and the two are never treated as the same thing.

3.2.6 Dynamic Availability Calculation and Assignment

The platform does not maintain a pre-generated table of appointment slots. Instead, availability is calculated dynamically at the moment a customer searches for or books an appointment. When a customer selects a service, the system checks, at the organization level, whether the business is open at the requested time and whether any special day affects that date. At the Worker level, it checks whether a given Worker is qualified to perform the requested service, whether their personal schedule covers the requested time, whether they have any approved block request that would conflict, whether they already have another appointment at that time, and, if the booking is for the current day, whether they are currently checked in. At the room level, it checks whether a compatible room exists for the service and whether that room is already occupied by another appointment at the requested time. Only when all of these conditions pass at every level is a given time offered to the customer as available.

This approach directly supports a "choose any available Worker" option, where the customer does not need to pick a specific person and the system instead searches across all qualifying Workers and compatible rooms in parallel and assigns whichever combination has capacity at the requested time. Calculating availability this way, rather than maintaining a separate slots table, keeps the schedules, block requests, rooms, and appointments as the single source of truth, avoids the risk of a stored slots table drifting out of sync with reality, and simplifies the overall data model. A booking itself only needs to record the chosen Worker, room, start time, and end time once confirmed.

3.2.7 Appointment Booking Flow

The appointment booking process is designed to be simple and to require as few steps as possible. A customer searches for an organization, selects a service, chooses either a specific Worker or "any available Worker," selects an available time computed dynamically as described above, and confirms the appointment. As part of confirming any new booking, the system validates that the appointment's scheduled end time does not fall after the organization's closing time for that day; if it would, that time is never offered to the customer in the first place. The resulting request enters the organization's queue as Pending until it is reviewed by the CRM, who approves it and confirms the assigned Worker and room. Once created, each appointment record stores the customer, the organization, the service, the assigned Worker, the assigned room, the scheduled start and end time, the actual start and end time once the service takes place, any notes, the current status, and payment information.

3.2.8 Appointment Lifecycle Management

Appointments move through a lifecycle that reflects what is actually happening rather than only what was originally planned. A booking starts out Pending while it awaits review by the CRM, becomes Confirmed once the CRM approves it, and moves to Checked In once the customer has arrived. From there it becomes In Progress once the Worker begins the service, and finally Completed once the Worker finishes and marks it as done. Along the way it may instead become Cancelled if it is called off before completion, Rejected if the CRM declines the request outright, or No Show if the customer never arrives.

Every appointment records both its scheduled start and end times and its actual start and end times, as these will not always match in practice. Recording the actual timestamps enables accurate reporting, operational analytics, and performance measurement.

When a Worker marks an appointment as Completed, the system records the actual completion time and compares it with the scheduled end time. If the Worker finishes more than five minutes earlier than scheduled, the platform notifies the next customer that their Worker has become available ahead of schedule and offers the earliest available appointment time. The customer can either accept or decline the offer, and the appointment is only moved earlier if the customer explicitly accepts, since it cannot be assumed they are nearby or able to arrive sooner.

If an appointment extends beyond its scheduled end time, the platform does not automatically delay every subsequent appointment, as a single automatic adjustment could create a cascading effect that disrupts the remainder of the day's schedule. Instead, once the delay exceeds five minutes, the CRM is notified of the predicted scheduling conflict along with the estimated impact on upcoming appointments. The CRM can then decide how best to resolve the situation, whether by delaying the next appointment, reassigning it to another available Worker, contacting the affected customer, or taking no action if the delay is expected to resolve quickly. This approach keeps scheduling decisions under human control while allowing the platform to proactively identify conflicts and recommend appropriate actions.

3.2.9 Appointment Calendar Management

The platform provides calendar views tailored to different users. Customers can view their upcoming and previous appointments along with each appointment's status and organization details. CRM users can view the organization's overall daily schedule, see appointments broken down by Worker, check available capacity, and review any approved block requests. Workers can view their own personal calendar of assigned appointments and blocked time. Across these views, the calendar supports color-coded appointments, appointment notes, and filtering by Worker.

3.2.10 Appointment History and Audit Trail

The platform keeps a full record of what happens to every appointment, but it does so as a separate event log rather than by duplicating the appointment's own data. Each history record is tied to a specific appointment and captures the action that was performed, the previous status, the new status, which person made the change, a timestamp, and an optional note. For example, if Appointment #123 moves from Pending to Confirmed, a corresponding history record is created showing that transition, who made it, and when. This keeps the appointment record itself focused on its current state while the history table exists purely to support auditing, transparency, and dispute resolution.

3.2.11 Worker Block Requests

Workers can submit requests to block off periods when they know in advance they will be unavailable, whether due to vacation, a personal appointment, a meeting, or any other planned reason. These requests are created by the Worker and must be approved by the CRM before they take effect; only after approval does a block request actually affect the availability the platform calculates for customers. This ensures that a Worker cannot unilaterally remove themselves from the schedule without the organization being aware and able to plan around it.

3.2.12 Payment Management

The platform connects payment information directly with appointments, supporting both cash and Visa/credit card payments. Each appointment tracks its payment method, payment status, and the time payment was completed; for example, a 100 USD consultation might show a Visa payment marked as Paid and completed at 2026-08-02 10:30 UTC. Payments themselves move through their own lifecycle, starting as Pending before payment is made, becoming Paid once payment succeeds, Failed if the attempt does not go through, or Refunded if the payment is later returned. A few business rules apply here as well: cash payments may remain unpaid until the appointment itself is completed, online payments require confirmation before they can be marked as paid.

3.2.13 Reviews and Ratings

Reviews help customers evaluate organizations, services and workers and strengthen trust across the marketplace, and customers can leave a review once their appointment has been completed. Each review includes a rating score, a comment, a creation date, and space for the organization to respond. To keep reviews meaningful, only customers with completed appointments may submit them, reviews are expected to reflect real customer experiences, and organizations may respond to a review but cannot remove it without approval from the platform.

3.2.14 Favorites

Customers can save services by an organization to their favorites list for faster booking in the future, adding or removing services as their preferences change and viewing their saved list whenever they want to quickly repeat a previous booking. This feature is a meaningful driver of customer retention and repeat appointments.

3.2.15 Notification System

Notifications belong to individual persons rather than to any specific role, and they keep both customers and organization staff informed as things happen, covering appointment confirmations, appointment reminders, appointment cancellations, payment confirmations, and organization announcements. Each notification records its receiving person, its message, its notification type, its read status, a created timestamp, and a read timestamp.

3.2.16 System Messages and Announcements

The platform also provides a channel of communication running from administrators to users, covering things like maintenance announcements, policy updates, and general platform announcements. Each system message carries its content, a message type, a creation date, and a read status.

3.2.17 Role-Based Access Control (RBAC)

The platform uses a predefined Role-Based Access Control model rather than one that organizations can configure themselves, and this control is enforced entirely within the application's internal logic rather than through editable database permission tables. Roles and their associated permissions are defined by the developer and shipped as part of the application, and organizations cannot create custom roles or modify existing permissions. The Super Admin can manage the platform as a whole, including organizations, users, and reports. The Organization Admin can manage their organization's profile, services, rooms, schedules, and staff, including assigning the CRM and Worker roles to individual people. The CRM can review and approve appointment requests, confirm Worker and room assignments, and review and approve Workers' block requests. The Worker can view the appointments assigned to them, request blocked time, and update the status of appointments they perform. The Customer can book appointments and manage their own profile, favorites, and reviews. Because permissions for each of these roles are enforced consistently by the application rather than configured per organization, authorization behavior stays the same across every organization on the platform, which simplifies administration, improves security, and guarantees predictable behavior throughout the system.

3.3 NON-FUNCTIONAL REQUIREMENTS

3.3.1 Security

The system must provide authentication and authorization, enforce role-based access control, handle user data securely, process payments securely, and protect against unauthorized access throughout.

3.3.2 Performance

The system should return any search results and check available appointment times to a customer quickly, handle multiple simultaneous bookings without issue, and prevent race conditions during the reservation process, particularly given that availability is calculated dynamically rather than read from a pre-generated table.

3.3.3 Reliability

The system must prevent double booking of both Workers and rooms, preserve appointment history accurately as an independent event log, maintain consistent payment records, and be able to recover gracefully from failures.

3.3.4 Scalability

The platform should be built to support growth all the way from a small number of individual organizations up to thousands of organizations and millions of customers, comfortably handling large volumes of appointments along the way, while remaining structured so that future extensions such as multi-branch organizations can be introduced without a fundamental redesign.